

ODOO FULL-STACK DEVELOPER PROGRAM

Training Presentation

Linux · Python · PostgreSQL · Odoo 17/18/19 · OWL · DevOps

Companion to the Master Training Plan · Professional Edition

AGENDA

What We Will Cover

01 Foundations

Linux CLI, Python for Odoo,
PostgreSQL administration

02 Odoo Core

Install, modules, fields, ORM,
views, security, data

03 Advanced

QWeb, OWL, POS, XML-RPC,
email & server actions

04 Production

Docker, VPS/Nginx/SSL, workers,
CI/CD & monitoring

PROGRAM VISION

From First Terminal Command to Production Odoo

- Build developers who can ship maintainable Odoo business modules
- Teach security and operations with the same rigor as features
- Cover Odoo 17 deeply, then map Odoo 18/19 deltas intentionally
- End with deployable artifacts—not only slide knowledge

OUTCOMES

Graduates Can...

Develop

Scaffold modules, design models/fields, override ORM safely, craft views

Secure

Groups, ACLs, record rules, Odoo 19 privilege awareness

Extend

QWeb PDFs, OWL widgets, POS customizations, XML-RPC integrations

Operate

Docker Compose, Nginx/SSL, systemd, worker sizing, backups

LEARNING JOURNEY

Six Progressive Phases

Phase	Modules	Exit Gate
1 Foundations	F1–F3	CLI + Python + SQL labs passed
2 Odoo Core	O1–O4	Custom module with CRUD logic
3 UX & Security	O5–O8	Secured app with list/form/search
4 Advanced	A1–A6	Report + OWL/POS + XML-RPC demo
5 Operations	D1–D4	HTTPS stack with workers tuned
6 Capstone	Projects	Reviewed production-ready delivery

PART ONE

Foundations

Linux · Python · PostgreSQL

MODULE F1

Linux OS & CLI Mastery

Navigate & Manage

pwd/cd/ls/tree · cp/mv/rm · permissions ·
users/sudo

Operate Services

systemctl · journalctl · processes · apt packages

Secure & Automate

SSH hardening · UFW · cron · env/.bashrc · disk
checks

Lab: project tree under /opt/training + sample systemd unit

MODULE F2

Python for Odoo Developers

- Core syntax → collections → control flow → functions
- OOP (inheritance, MRO, dunder) mapped to Odoo model thinking
- Functional tools, decorators, errors, generators, packages
- Tooling: venv, pip, PEP8, VS Code/PyCharm, debugging

Lab Cap: CLI contact manager with classes + persistence

MODULE F3

PostgreSQL Administration

Setup

Install on Ubuntu LTS · psql · roles · grants ·
pg_hba/conf

SQL Fluency

DDL · CRUD · joins · aggregations · constraints

Ops

Indexes · EXPLAIN · pg_dump/restore · locks &
sessions

PART TWO

Odoo Core Development

Install → Modules → Fields → ORM → Views → Security

MODULE 01

Installation, CLI & Configuration

- Odoo 17 on Ubuntu 22.04 · Odoo 19 preview on Ubuntu 24.04
- odoo-bin startup · configuration file · PyCharm run configs
- Database manager URLs · disable DB selector · master password
- Install/upgrade modules via CLI · shell · logs · migration upgrade code

MODULE 02

Modules, Manifest & Addons Paths

Scaffold

Create modules with scaffold; understand
_name/_table

Manifest

Metadata, depends, data/demo ordering,
application flag

Paths

Multi addons-path · App Store layouts ·
auto-install patterns

MODULE 03

Fields & Model Design

Category	Field Types Practiced
Scalar	Char · Text · Html · Boolean · Selection · Integer · Float · Json
Temporal	Date · Datetime
Relational	Many2one · One2many · Many2many · Reference · Related
Advanced	Compute · Monetary · Binary · Image

MODULE 04

ORM Methods Deep Dive

Lifecycle

create · write · copy · unlink ·
custom methods · exists

Query APIs

search/domains · read* ·
name_create · default_get ·
search_fetch

Recordsets

ensure_one · filtered · mapped ·
sorted · grouped

Special Cmds

[0]..[6] relational commands ·
active/archive patterns

MODULE 05

Views for Real Business UX

- List: editable, decorations, mass edit, headers/buttons, aggregates, handle widget
- Search: panel, dynamic datetime filters, default filters
- Form: sheet/header/group/notebook/buttons/statusbar
- Kanban + Graph/Pivot/Cohort/Gantt · attrs → invisible/readonly/required migration

MODULE 06

Security — The Non-Negotiable Layer

Groups

Categories · parent/child ·
implied_ids · combobox layouts

Access

ir.model.access via CSV and XML

Record Rules

Teacher/Student/Admin scenarios ·
inheritance

Odoo 19

Privilege model changes · groups
on fields/views/actions

MODULES 07-08

Data Loading & View Inheritance

XML & CSV Data

Records, eval, images, M2O/M2M/O2M,
binaries; when to choose XML vs CSV

Inheritance

field tag · xpath · positions
after/before/inside/replace/attributes

Extras

URL actions · ir.sequence auto-numbering ·
upgrade-safe xpath habits

PART THREE

Advanced Odoo

QWeb · OWL · POS · APIs · Automation

MODULE A1

QWeb PDF Reports

- t-field / t-esc / t-out · conditionals · loops · variables · t-att/t-attf
- Template inheritance · paper formats · barcodes/QR · static & dynamic images
- Lab: enrollment confirmation / student ID PDF

MODULE A2

OWL Framework

Components

Lifecycle · useState/useRef · events · props

Templates

t-if · t-foreach · t-model · custom field widgets

Integration

patch() · rpc service · custom
Kanban/Dashboard views

MODULE A3

POS Customization with OWL

- Custom buttons · click handlers · hide standard controls
- Popups · JS translations · RPC fetch/render · clear order lines
- Lab: Manager Note popup persisted on pos.order

MODULES A4–A6

Integrations & Automation

XML-RPC

Auth · export · CRUD · Postman verification ·
least-privilege users

Email

Templates · send flows tied to business events

Server Actions

Use cases · UI triggers · safe Python practices

PART FOUR

DevOps & Production

Docker · VPS · Nginx · Performance · CI/CD

MODULE D1

Docker & Compose for Odoo

- Images/containers/volumes/networks · custom Dockerfiles
- Compose: Odoo + PostgreSQL · .env · upgrade without data loss
- CI/CD primer with GitHub Actions for module checks

MODULE D2

VPS Deployment Blueprint

Layer	Action
Security	SSH hardening · UFW · least privilege odoo user
Layout	/opt/odoo · logs · custom addons
Proxy/TLS	Nginx 80/443 → 8069 · longpolling 8072 · Certbot
Runtime	systemd odoo.service · cron DB+filestore backups

MODULE D3

Workers & Performance

Sizing

Workers from CPU/RAM · memory & time limits

Data Layer

max_connections · missing indexes ·
pg_stat_statements

Scale Out

Longpolling · Redis sessions · Nginx LB ·
Prometheus/Grafana

DELIVERY MODEL

Time & Format Options

120+	12–16	6–8	3
Guided learning hours	Weeks professional track	Weeks intensive bootcamp	Capstone project choices

CAPSTONES

Choose a Path to Demonstrate Mastery

A · Institute Manager

Students, courses, enrollment, fees, security roles, QWeb PDF, email automation

B · Retail POS Helper

OWL POS button/popup/RPC note on orders + manager ACL + optional XML-RPC export

C · Ops Excellence

systemd + Nginx/TLS + backups + worker worksheet + restore rehearsal

ASSESSMENT

How Competence Is Measured

Component	Weight	Evidence
Module Labs	40%	Working code + screenshots
Knowledge Checks	15%	ORM · security · CLI · SQL quizzes
Mini-Projects	20%	Integrated feature slices
Final Capstone	25%	Deployable module + docs + demo

12-WEEK SNAPSHOT

Professional Track at a Glance

Weeks	Focus
1–4	Linux · Python · PostgreSQL foundations
5–7	Odoo install · modules · fields · ORM
8–9	Views · inheritance · security (+ Odoo 19 notes)
10–11	QWeb · OWL/POS · XML-RPC · email/actions
12	Docker · VPS/Nginx · workers · capstone defense

FACILITATION PRINCIPLES

How We Teach

- Outcomes first — every session names an exit gate
- Security before spectacle — ACLs precede fancy widgets
- Version honesty — Odoo 17 core, Odoo 19 deltas called out explicitly
- Artifacts over attendance — labs must run on a reviewer's machine

NEXT STEP

Open the Training Plan and Start Module F1

Document: [Odoo_FullStack_Developer_Training_Plan.pdf](#)

Weblearns Academy · Odoo Full-Stack Developer Program